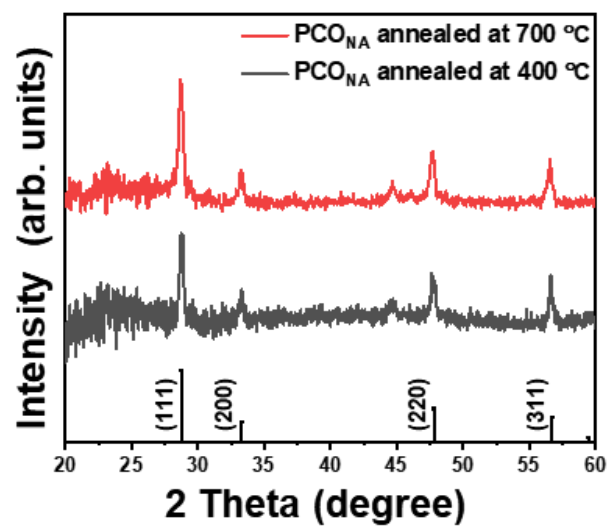
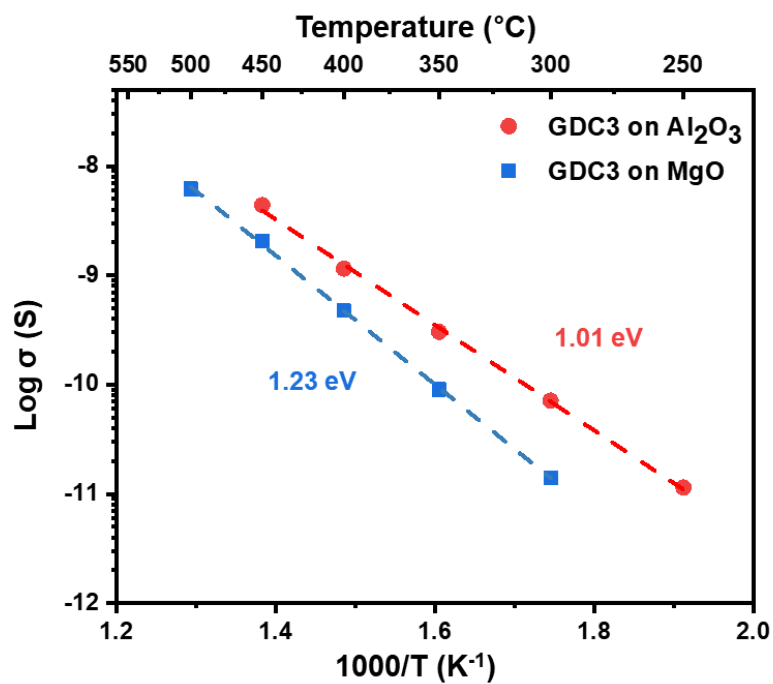
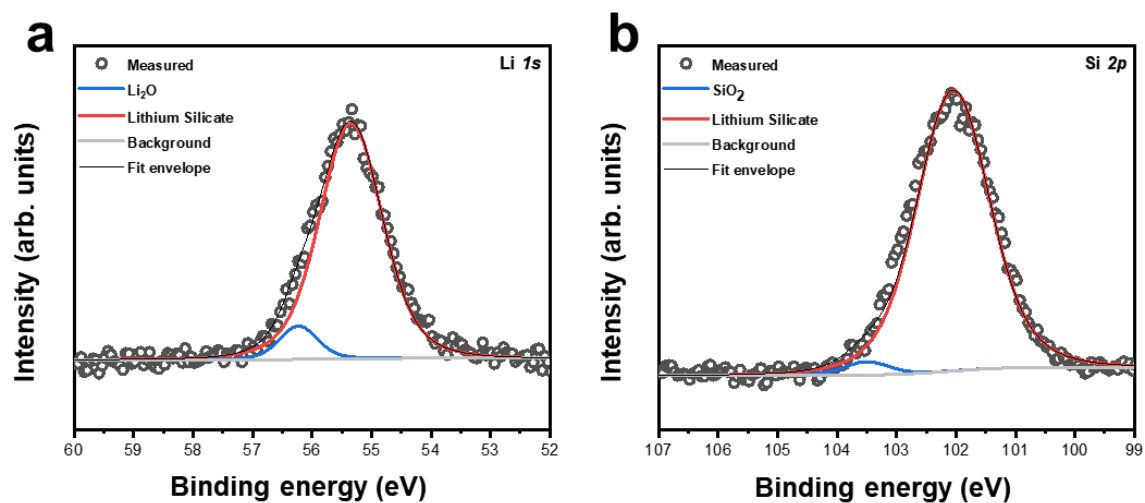




Supplementary Fig. 9 XRD spectra of PCO_{NA} fabricated on Al₂O₃ substrate annealed at 400 °C and 700 °C, respectively. Black vertical lines at bottom of figure indicate the dominant facets of the cerium oxide crystal structure.



Supplementary Fig. 10 Arrhenius plots of in-plane conductivity of 3 at% Gd-doped ceria (GDC3) nanowire arrays fabricated on Al_2O_3 and MgO substrates annealed at 400 $^{\circ}\text{C}$ along with their respective activation energies.



Supplementary Fig. 11 XPS spectra of PCO_{NA} after sequential infiltration with Li-species following Si-species, showing the different binding energy regions of Li $1s$ and Si $2p$ spectra, respectively.